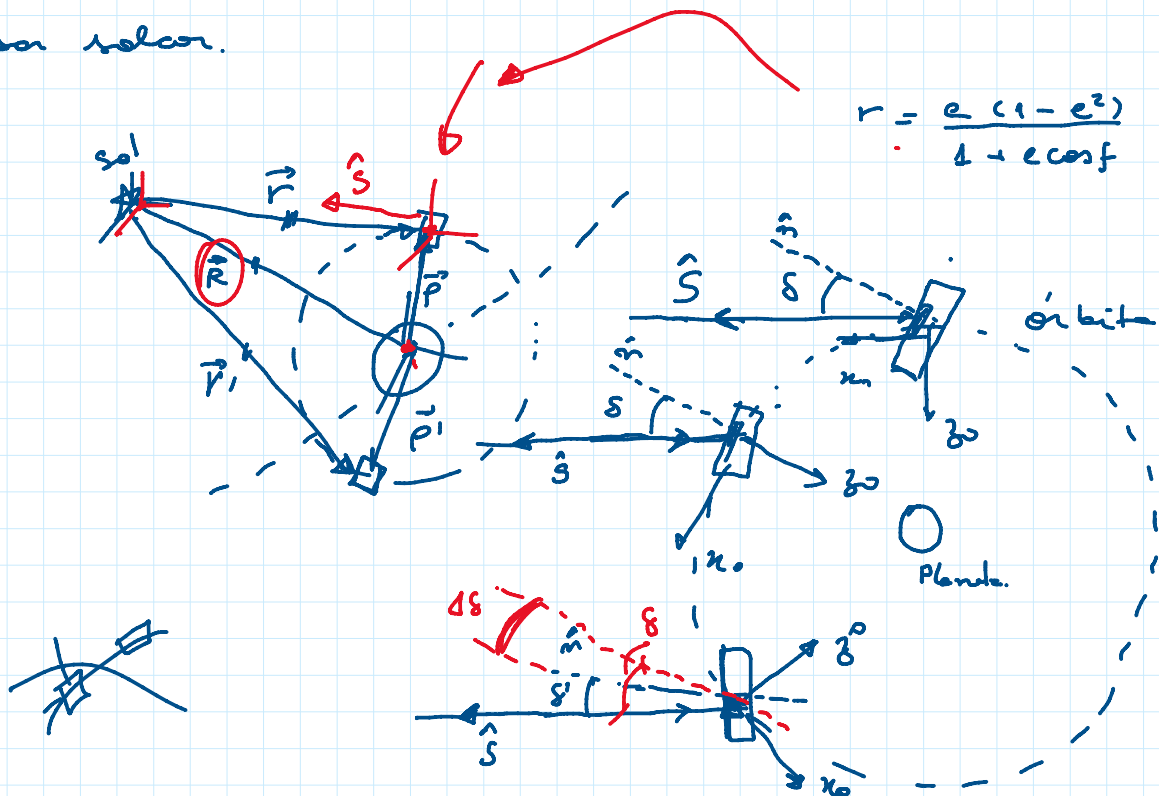
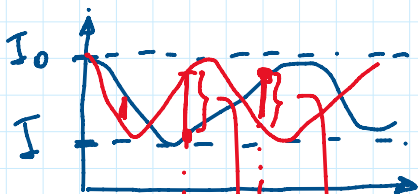
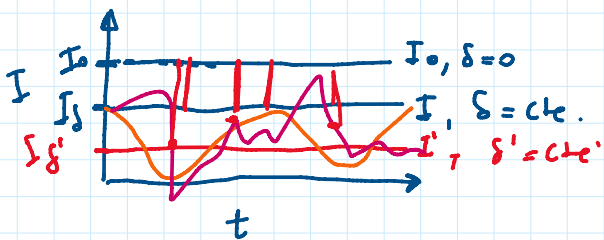
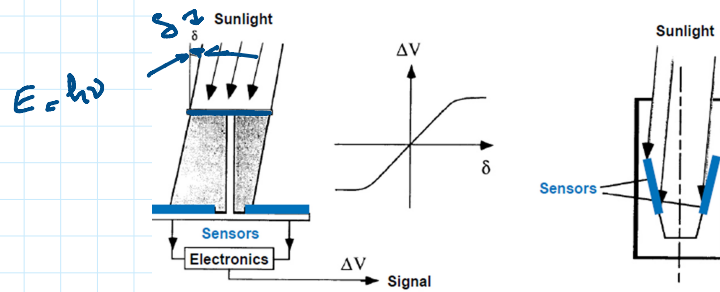


Sensor vector.



$$r = \frac{a(1-e^2)}{1+e\cos f}$$



$$\Delta I \rightarrow \Delta \delta$$

$$\begin{cases} I(\delta) = I_0 \cos(\delta) \\ \delta = 0 \\ I(\delta) = I_0 \end{cases}$$

$$\begin{cases} \delta = cte \\ I(\delta) = I_0 \cos(\delta) = cte. \end{cases}$$



Quatérnio:

$$\bar{q} = u + x\hat{i} + y\hat{j} + z\hat{k}$$

$$\bar{q} = q_1 + q_2 + q_3 + q_4$$

$$\bar{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} 0,1041 \\ -0,2374 \\ -0,5480 \\ 0,7953 \end{bmatrix}$$

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1$$

confirmando!

Exemplo

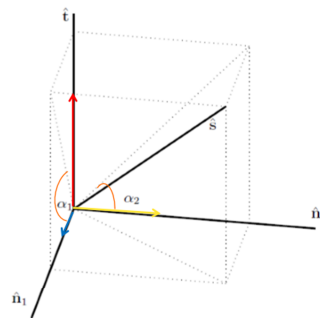
$H \rightarrow C$

Suponha que um sensor solar seja configurado como mostrado na figura a seguir, com $\alpha_1 = 0,9501$ rad, $\alpha_2 = 0,2311$ rad e a orientação do sistema fixo no sensor solar com respeito ao sistema fixo no corpo é descrito pelo quatérnio:

$q = [0,1041 \ -0,2374 \ -0,5480 \ 0,7953]^T$, $q_1 = 0,1041$, $q_2 = -0,2374$, $q_3 = -0,5480$ e $q_4 = 0,7953$.

determine:

- s_s^*
 - Magnitude de s_s^*
 - s_s
 - s_b
 - S_i (para $\theta_1 = 30^\circ$, $\theta_2 = 15^\circ$, $\theta_3 = 51^\circ$)
- Use as matrizes do slide 17.



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$$a) \quad s_s^* = \begin{bmatrix} 1 & \frac{\tan \alpha_1}{\tan \alpha_2} & \tan \alpha_1 \end{bmatrix}^T = \begin{bmatrix} 1 \\ \frac{\tan \alpha_1}{\tan \alpha_2} \\ \tan \alpha_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5,9441 \\ 1,3987 \end{bmatrix}$$

$$b) \quad \|s_s^*\| = \sqrt{s_s^{*T} s_s^*} = ?$$

$$s_s^{*T} s_s^* = \begin{bmatrix} 1 & 5,9441 & 1,3987 \end{bmatrix} \begin{bmatrix} 1 \\ 5,9441 \\ 1,3987 \end{bmatrix} = 1^2 + 5,9441^2 + 1,3987^2$$

$$s_s^{*T} s_s^* = 38,2887$$

$$\|s_s^*\| = 6,1878$$

$$c) \quad s_s = \frac{s_s^*}{\|s_s^*\|} = \frac{\begin{bmatrix} 1 & 5,9441 & 1,3987 \end{bmatrix}^T}{6,1878} = \begin{bmatrix} 0,1616 & 0,9606 & 0,2260 \end{bmatrix}^T$$

$$s_s = \begin{bmatrix} 0,1616 \\ 0,9606 \\ 0,2260 \end{bmatrix}$$

$$d) \quad s_b = R^{bs} s_s \quad R^{bs} = (\underline{q_4^2} - \underline{q^T} \bullet \underline{q}) \mathbf{1} + 2 \underline{q} \underline{q^T} - 2 \underline{q_4} \underline{q}^\times$$

$$q_4^2 = 0,7953^2 = 0,6325$$

$$\mathbf{q}^T \cdot \mathbf{q} = [q_1 \quad q_2 \quad q_3] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = q_1^2 + q_2^2 + q_3^2 = \begin{bmatrix} 0,1041 & -0,2374 & -0,5480 \end{bmatrix} \begin{bmatrix} 0,1041 \\ -0,2374 \\ -0,5480 \end{bmatrix}$$

$$= 0,3675$$

$$2\mathbf{q}\mathbf{q}^T = \begin{bmatrix} q_1^2 & q_1 q_2 & q_1 q_3 \\ q_2 q_1 & q_2^2 & q_2 q_3 \\ q_3 q_1 & q_3 q_2 & q_3^2 \end{bmatrix} = \begin{bmatrix} 0,0216 & -0,0494 & -0,1140 \\ 0,0108 & -0,0247 & -0,0570 \\ -0,0494 & 0,1128 & 0,2600 \\ -0,0247 & 0,0564 & 0,1300 \\ -0,1140 & 0,2600 & 0,6000 \\ -0,0570 & 0,1300 & 0,3000 \end{bmatrix}$$

$$2q_4 = 1,5906$$

$$\mathbf{q}^\times = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0,5480 & -0,2374 \\ -0,5480 & 0 & -0,1041 \\ 0,2374 & 0,1041 & 0 \end{bmatrix}$$

$$2q_4 \mathbf{q}^\times = \begin{bmatrix} 0 & 0,8716 & -0,3776 \\ -0,8716 & 0 & -0,1656 \\ 0,3776 & 0,1656 & 0 \end{bmatrix}$$

$$\mathbf{R}^{bs} = 0,2650 \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{1}} + \begin{bmatrix} 0,0216 & -0,0494 & -0,1140 \\ -0,0494 & 0,1128 & 0,2600 \\ -0,1140 & 0,2600 & 0,6000 \end{bmatrix} - \begin{bmatrix} 0 & 0,8716 & -0,3776 \\ -0,8716 & 0 & -0,1656 \\ 0,3776 & 0,1656 & 0 \end{bmatrix}$$

$$\mathbf{R}^{bs} = \begin{bmatrix} 0,2867 & -0,9211 & 0,2635 \\ 0,8222 & 0,3777 & 0,4258 \\ -0,4917 & 0,0946 & 0,8656 \end{bmatrix}$$

$$\mathbf{S}_b = \mathbf{R}^{bs} \mathbf{S}_s = \begin{bmatrix} 0,2867 & -0,9211 & 0,2635 \\ 0,8222 & 0,3777 & 0,4258 \\ -0,4917 & 0,0946 & 0,8656 \end{bmatrix} \begin{bmatrix} 0,1616 \\ 0,9606 \\ 0,2260 \end{bmatrix}$$

$$\mathbf{S}_b = \begin{bmatrix} -0,7789 \\ 0,5920 \\ 0,2071 \end{bmatrix} = \begin{bmatrix} -0,7789 & 0,5920 & 0,2071 \end{bmatrix}^T$$

$$e) \quad S_J = ?$$

$$S_b = R^{bi} S_i \rightarrow S_i = (R^{bi})^{-1} S_b.$$

$$R^{bi} = ?$$

$$R^{bi} = R_1(\theta_3) R_2(\theta_2) R_3(\theta_1)$$

$$R_3(\theta_1) = \begin{bmatrix} \cos \theta_1 & \sin \theta_1 & 0 \\ -\sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R_2(\theta_2) = \begin{bmatrix} \cos \theta_2 & 0 & -\sin \theta_2 \\ 0 & 1 & 0 \\ \sin \theta_2 & 0 & \cos \theta_2 \end{bmatrix} \quad R_1(\theta_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_3 & \sin \theta_3 \\ 0 & -\sin \theta_3 & \cos \theta_3 \end{bmatrix}$$

$$(R^{bi})^{-1} = \begin{bmatrix} 0,8365 & -0,1405 & 0,5296 \\ 0,4830 & 0,6456 & -0,5964 \\ -0,2550 & 0,7507 & 0,6079 \end{bmatrix}$$

$$S_i = [-0,6250 \quad -0,1165 \quad 0,7418]^T$$